

qPOTS: A Python Package for Sample-efficient Batch Constrained Multiobjective Bayesian Optimization

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Summary

Designing engineered systems involve optimizing the system for some performance objective under several design constraints. In practice, though, there are typically more than one objective to meet and the objectives may conflict with each other. For instance, an aircraft designer may need to reduce the aerodynamic drag on the aircraft while ensuring passenger safety. Aerodynamic efficiency typically leads to slender aircraft cross-sections which can, however, compromise the safety of the aircraft under gusty conditions. Similar tradeoffs occur in chemical process optimization, materials discovery, and machine learning hyperparameter optimization. Optimizing simultaneously for more than one objective is called multiobjective optimization, and the solution to such problems is no longer a single point but a set of points that offer a pragmatic compromise between the objectives.

qPOTS is a Python package that solves constrained multiobjective optimization when there is no derivative information, and when every query of the objectives and constraints is expensive and thus necessitating “sample efficiency” (i.e., solving the problem with as few evaluations of objectives and constraints as possible). qPOTS is a Bayesian optimization (Frazier, 2018, p. garnett2023bayesian) approach, and implements the batch Pareto optimal Thompson sampling (qPOTS) method (Renganathan & Carlson, 2025; Thompson, 1933). The package also includes qPOTS-Decoupled, which supports decoupled objective and constraint evaluations – that is when objective and constraints are evaluated by running computer models that can be evaluated independently, leveraging independent resources, the algorithm achieves further sample efficiency by evaluating only a subset of objectives and constraints during each design iteration (Renganathan et al., 2026).

qPOTS is released under the GNU General Public License v3.0 and is built on widely used scientific Python libraries including PyTorch, GPyTorch, BoTorch, and Pymoo (Balandat et al., 2020; Blank & Deb, 2020; Gardner et al., 2018; Paszke et al., 2019).

Statement of need

The solution to a multiobjective optimization problem is called the “Pareto set”, and their image in the objective space is called the “Pareto frontier”. Computing the Pareto set/frontier is nontrivial. When derivative information is available and the objectives and constraints are smooth and not expensive to evaluate, one may use gradient-based optimization techniques over several scalarized versions of the objectives Fliege et al. (2009). However, in real-world problems, good quality derivatives are rarely available, objective/constraint smoothness is not guaranteed, and the Pareto frontier may have a complex topology (disconnected, nonconvex, etc.). In such situations, multiobjective Bayesian optimization (MOBO) serves as a suitable alternative. MOBO learns global surrogate models for the objective and constraints, and leverages Bayesian decision theory to generate a sequence of iterates that balance exploration

and exploitation of the design space. This involves constructing an “acquisition” function that quantifies the utility of a candidate iterate, and solving an “inner” optimization problem to determine the iterates. This inner optimization depends only on the surrogate model and is independent of the expensive objectives and constraints.

MOBO is an active area of research; there are several existing works that innovate on the Bayesian decision-theoretic acquisition function Daulton et al. (2020), or use scalarization to fit into (single objective) BO (Knowles, 2006), or modify existing approaches to promote diversity in the iterates (Konakovic Lukovic et al., 2020), among others. However, existing methods have one or more of the following limitations. (1) The inner acquisition function optimization is still nontrivial due to non-convexity and/or stochasticity - this ultimately impacts the accuracy of the predicted Pareto set/frontier. (2) They don’t support batch sampling (generating a batch of iterates). (3) They cannot handle nonlinear constraints. See (Renganathan & Carlson, 2025) for further details. qPOTS was introduced to address all of these limitations, with improved sample efficiency and solution accuracy being our overarching goal.

Evolutionary algorithms (Deb et al., 2002) are quite popular for derivative-free multiobjective optimization. However, they rely on generating large populations of the objectives that must be mutated and crossed-over multiple times. This, although leads to accurate solutions, can penalize sample efficiency. qPOTS innovatively combines the sample efficiency of BO with the accuracy of evolutionary algorithms. By fitting Gaussian process (Rasmussen & Williams, 2006) surrogates for the objectives and constraints, qPOTS replaces the conventional acquisition function optimization in MOBO with a cheap inner multiobjective optimization over the GP posterior sample paths, leveraging evolutionary algorithms. Nonlinear constraints are seamlessly handled as part of the Gaussian process Thompson sampling (Renganathan & Carlson, 2025).

It should be mentioned that Thompson sampling based approaches, similar to qPOTS, already exist for MOBO – TS-EMO and TS-TCH Paria et al. (2020). The crucial difference between qPOTS and TS-EMO is that TS-EMO generates GP posterior Thompson samples, but still has to compute the maximum hypervolume improvement over them to choose the candidate – this inner optimization problem can be difficult to solve (Renganathan & Carlson, 2025). Additionally TS-EMO is unable to handle constraints. TS-TCH scalarizes the Thompson samples to leverage single-objective acquisition functions. This inherits the limitations scalarized approaches in computing disconnected and nonconvex Pareto frontiers.

A further opportunity arises when the objectives and constraints can be evaluated asynchronously using independent compute resources. For example, in aerodynamic design, objectives may be aircraft performance at different operating conditions that may be evaluated separately. In such situations, if the objectives and constraints are correlated with each other, MOBO can proceed with only a subset of objective and constraint evaluation at each iterate. The qPOTS library includes qPOTS-Decoupled (Renganathan et al., 2026) to address this, which combines the qPOTS posterior sampling workflow with a multitask Gaussian process (MTGP) (Bonilla et al., 2007) that jointly learns all objectives and constraints. Crucially, MTGPs learn the correlations across the objectives and constraints which can be exploited to determine which iterates warrant a decoupled evaluation. It should be noted that decoupled MOBO methods already exist - PESMOC and HVKG can progress with partial information (Daulton et al., 2023; Garrido-Merchán & Hernández-Lobato, 2019). However, existing methods depend on evaluation cost difference across the objectives and constraints to determine decoupling. On the other hand, qPOTS-Decoupled can decouple evaluations under uniform evaluation costs; crucially, evaluation cost difference will further improve sample efficiency of qPOTS-Decoupled.

qPOTS builds on GPyTorch (Gardner et al., 2018) and existing GP and MTGP wrappers in BoTorch (Balandat et al., 2020) for surrogate modeling, Pymoo (Blank & Deb, 2020) for evolutionary algorithms, and fully supports GPU acceleration. Even though qPOTS builds on these existing libraries, a need for a standalone library is compelling for the following reasons. First, qPOTS is anchored on Pareto optimal Thompson sampling – this makes it compatible with probabilistic models beyond GPs such as Bayesian neural networks and variational autoencoders.

94 Therefore, a future direction independent of GP models is possible. Second, generative models
95 may be integrated into the framework to learn the distribution of the Pareto set $\mathcal{X}^*$ in
96 Thompson sampling. Third, qPOTS-Decoupled can be potentially applied to other problems
97 such as optimal experiment design and sensor placement, where optimal selection of a subset
98 from a correlated set of tasks is of interest. Therefore, a standalone library will promote wider
99 usage, contributions, and expansions that might not be possible when integrated into another
100 library.

101 The target audience is researchers and engineers who need a sample-efficient derivative-free
102 multiobjective optimization software in Python. qPOTS is designed for method developers
103 benchmarking against standard acquisition strategies, and for application researchers optimizing
104 expensive custom black-box functions under limited evaluation budgets.

105 State of the field

106 While multiobjective optimization can be solved via a variety of approaches including gradient-
107 based approaches and evolutionary algorithms, we only review libraries relevant to MOBO
108 here. BoTorch provides a PyTorch-based platform for Bayesian optimization (including MOBO)
109 and already implements MOBO acquisition functions such as qNEHVI, qNParEGO, and their
110 logarithmic variants (Ament et al., 2023; Balandat et al., 2020; Daulton et al., 2020), entropy-
111 search acquisitions (PESMOC, MESMO, and JESMO) (Belakaria et al., 2019; Garrido-Merchán
112 & Hernández-Lobato, 2019; Tu et al., 2022), and hypervolume knowledge gradient (Daulton
113 et al., 2023). Trieste provides a modular TensorFlow-based Bayesian-optimization loop with
114 multiobjective and constrained workflows (Picheny et al., 2023). TS-EMO (Bradford et al.,
115 2018) is another Thompson sampling based MOBO algorithm (similar to qPOTS) that is based
116 on MATLAB. qPOTS has a clear methodological distinction from these libraries as explained in
117 the previous section and in more detail in (Renganathan & Carlson, 2025), while also having a
118 demonstrated better empirical performance in comparison (Renganathan & Carlson, 2025).

119 The ability of qPOTS to go beyond GP models (as is standard in MOBO), to interface with
120 generative modeling libraries, and extend to other applications including optimal experiment
121 design and sensor placement, makes it attractive as a standalone package that users can build
122 on.

123 Software design

124 qPOTS is an algorithm-specific coordination layer, not another BO library, evolutionary optimizer,
125 or workflow engine. Each iteration must keep physical coordinates, normalized tensors,
126 posterior state, Pymoo populations, and, in decoupled runs, task identities consistent. The
127 package centralizes the invariant fit $\rightarrow$ posterior-path search $\rightarrow$ batch/task selection
128 $\rightarrow$ evaluation $\rightarrow$ update transition in a runner.

129 The runner exposes model and acquisition objects and iteration callbacks. These seams support
130 surrogate ablations, alternative candidate policies, and external simulator hooks without copying
131 the optimization loop. Their cost is indirection and a larger API surface; concrete defaults,
132 validated configuration, typed results, and fixed control flow contain it. Routine use calls
133 `run()`, while `step()` exposes each iteration to external orchestration.

134 Coupled and decoupled modes can be run with MTGPs and share one state representation:
135 task identifiers record requested outputs and unevaluated outputs become missing observations.
136 This avoids parallel implementations, but requires stricter validation and a missing-data capable
137 surrogate. It also makes qPOTS more than a BoTorch acquisition function: the package
138 coordinates posterior path construction, Pareto set search, batch and output selection, and
139 partial observation updates across library boundaries.

140 BoTorch and GPyTorch supply tensor-native modeling, posterior sampling, and baseline

acquisition components (Balandat et al., 2020; Gardner et al., 2018). Reimplementing them would add numerical maintenance without distinguishing qPOTS. Pymoo is retained because its vectorized bounded-problem interface, seeded execution, nondominated-set result, and callbacks fit the inner multiobjective optimization needs well (Blank & Deb, 2020). jMetalPy, pagmo/pygmo, and DEAP are credible alternatives (Benítez-Hidalgo et al., 2019; Biscani & Izzo, 2020; Fortin et al., 2012). The cost is an extra dependency and a NumPy–PyTorch/CPU boundary; isolating the adapter localizes replacement. We did not use a file/DAG orchestrator such as Snakemake (Köster & Rahmann, 2012) because each next design – and sometimes output task – depends on a newly fitted in-memory posterior. The corresponding cost is that QPOTSRunner does not provide file-based checkpointing, scheduling, or provenance; external systems can invoke `step()` and consume callbacks.

Research impact statement

qPOTS is the reference implementation for the AISTATS 2025 paper introducing Pareto optimal Thompson sampling for batch multiobjective Bayesian optimization (Renganathan & Carlson, 2025). That paper evaluates the method on synthetic and real-world benchmarks against analytical, entropy-search, Thompson-sampling, and random baselines. The software has also supported an aerodynamic design study of the NASA Common Research Model with open reproducibility materials (Carlson & Renganathan, 2026). These are demonstrated uses of the method and its research code lineage.

Independent evaluation is beginning to emerge: SPREAD includes qPOTS in a numerical comparison of MOBO methods (Hotegni & Peitz, 2025). Other publications cite qPOTS while discussing generative molecular design, sustainable process systems, class-imbalance learning, and many-objective optimization (Jiang et al., 2026; Kudva et al., 2026, Ch. 3; Muthyala et al., 2026; Wang et al., 2025). We treat those citations as evidence of both awareness of qPOTS and as a state-of-the-art benchmark in the MOBO community.

AI usage disclosure

Anthropic Claude assisted with code refactoring and README structure before submission. OpenAI Codex assisted during review revisions with drafting tests, examples, and documentation. Generative AI was not used to design the qPOTS or qPOTS-Decoupled algorithms or to generate benchmark results. The authors reviewed every change, reran the software tests and documented examples, checked citations against publication records, and remain responsible for the manuscript and software.

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